Based on the available search results, I cannot provide a definitive, fully optimized formulation for your specific application. The search results do not contain a dedicated study on NR/EPDM hydro-seal compounds reinforced with **tetrapod-shaped ZnO whiskers (T-ZnOw)** and **nano-silica** for hydro-power plant applications.

However, I can synthesize relevant data from adjacent research areas to propose a starting formulation and processing parameters that you can use as a baseline for further experimental optimization.

□ Proposed Baseline Formulation

The table below outlines a suggested starting point, drawing on T-ZnOw usage in EPDM seals and conventional NR/EPDM ratios .

| Component | Suggested Loading (phr) | Rationale / Source Data |

| :--- | :--- | :--- |

| **NR / EPDM Blend** | 70 / 30 | A common ratio balancing NR's mechanical strength with EPDM's aging resistance . |

| **T-ZnOw** | **1.0 – 2.0** | A patent for EPDM gaskets uses **0.5–1.5 phr** . Another source for water-resistant seals uses **6–15 phr** . Start at the lower end for reinforcement, or higher for specific functional properties (e.g., water/oil resistance). |

| **Nano-Silica** | **10 – 20** | A patent for EPDM seals uses **1–3 phr** of fumed silica . For significant mechanical reinforcement, higher loadings are typical. |

| **Zinc Oxide (Activator)** | 5.0 | Standard curing activator for sulfur systems . |

| **Stearic Acid** | 1.0 – 2.0 | Co-activator with ZnO . |

| **Carbon Black (e.g., N330)** | 30 – 50 | Primary reinforcing filler. |

| **Process Oil** | 5 – 10 | To aid processing and adjust hardness. |

| **Sulfur** | 1.5 – 2.0 | Curing agent. |

| **Accelerator (e.g., TBBS, CZ)** | 1.0 – 1.5 | Vulcanization accelerator . |

Note on T-ZnOw: While it can enhance mechanical properties, its primary reported benefit in EPDM seals is often as an anti-mildew agent without negatively affecting cure rate or product quality .

□ Suggested Vulcanization Parameters

The optimal cure temperature and time depend on your specific compound and the desired state of cure.

* **Temperature:** A vulcanization temperature of **150°C to 170°C** is standard for many NR/EPDM and EPDM seal formulations .

* **Time:** You must determine the **optimum cure time (t90)** using a rheometer (MDR or ODR) for your specific compound. Typical press cure times range from **15 to 30 minutes**. A specific patent example for an EPDM seal cures at **170°C for 20 minutes** .

□ Recommended Path Forward

To develop a truly optimized formulation, I recommend the following experimental steps:

1. **Conduct a Design of Experiments (DoE):** Vary the T-ZnOw and nano-silica loadings systematically. This will allow you to model their effects on critical properties like tensile strength, compression set, and low-temperature elasticity.

2. **Determine Cure Kinetics:** Use a Moving Die Rheometer (MDR) to generate the cure curve for your chosen formulation. This will give you the precise **t90** and other kinetic parameters needed to set the optimal cure time.

3. **Test Critical Performance Properties:** Your evaluation should include:

    * **Water Resistance:** Water contact angle measurements and water absorption tests.

* **Cold Resistance:** Low-temperature compression set (e.g., ASTM D1229) and TR-10 testing to determine low-temperature flexibility .

* **Mechanical Properties:** Tensile strength and elongation at break, both before and after aging.

I hope this provides a solid foundation for your development work. If you can share more specific performance targets (e.g., target hardness, compression set limits), I can help refine the initial formulation further.